

PERSONAL DETAILS

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📍 13100 Aix-en-Provence, France
🌐 lg-space.github.io/portfolio/en.html
20 years old

IT SKILLS

Software
Microsoft Office (Word, Excel, PowerPoint), Canva, L^AT_EX, Git/GitHub, SolidWorks, CATIA, Granta

Programming languages

Python	●●●●●
C++	●●●●●
MATLAB/Octave	●●●●●
SQL	●●●●●
Fortran	●●●●●
HTML/CSS	●●●●●

Self-assessed, from basic to advanced

LANGUAGES

French: mother tongue
English: upper-intermediate (B2), Cambridge English certificate
Spanish: intermediate (B1), DELE certificate

SOFT SKILLS

Teamwork
Project management
Time management
Giving presentations
Written communication
Rigour and analytical thinking

INTERESTS

Water polo: sports-study programme at school; team spirit and discipline.
Sailing and skiing
Mountain biking, golf, horse riding
Krav maga
Trombone

REFERENCES

Mr Cédric Galusinski
Head of the MOCA track
SeaTech, University of Toulon
✉ cedric.galusinski@univ-tln.fr
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Mr Stéphane Castells
Regional Manager
Géoterria
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Lysandre GOUDON

Engineering student (M.Eng.) in Computational Fluid Dynamics

PROFILE

Second-year engineering student at SeaTech, specialising in numerical modelling and fluid mechanics applied to the marine sector. Seeking an internship in **CFD, numerical simulation or ocean engineering**. Strong in mathematics, programming and data analysis.

EDUCATION AND QUALIFICATIONS

2025 – 2028 (expected) Toulon, France 
SeaTech, School of Engineering, University of Toulon
• Preparing an **M.Eng. in Computational Fluid Dynamics**: MOCA track (Modelling and Computation of Fluids and Structures)
• **Subjects:** numerical analysis, partial differential equations (PDEs), fluid mechanics, wave physics, signal analysis, databases
• **Minor in Security and Defence:** geostrategy, infrastructure resilience, dual-use (maritime and defence) technologies

2023 – 2025 Lyon, France 
Classes Préparatoires aux Grandes Écoles, Lycée Aux Lazaristes
Intensive pre-engineering programme for entrance into elite engineering schools (PC track)
• **Subjects:** physics, mathematics, chemistry
• Carried out a supervised research project (TIPE) on supercapacitor energy density

2022 – 2023 Aix-en-Provence, France
ACESUP: early preparation for CPGE
• Intensive pre-course for scientific preparatory classes

2020 – 2023 Aix-en-Provence, France
Lycée Sainte-Catherine-de-Sienne, A-Levels equivalent
• French *Baccalauréat* (general), **with high honors**
• Subjects: mathematics, physics and chemistry; European section

WORK EXPERIENCE

Work placement
2019 – 2026 Castagniccia, Corsica
Project assistant: GAL Castagniccia Mare à Monti
Local Action Group (LAG) funded by the European LEADER programme
• Worked on the renovation of old mountain houses to create tourist accommodation
• Contributed to a rural development project supported by European funding

Internship
Year 1 placement
Geotechnical intern: Géoterria (geotechnical consultancy)
• Studied and characterised soils; analysed building foundations
• Applied continuum mechanics principles to real sites

Projects
Year 1 team project
Mobile oceanographic probe: Astrolabe Expeditions
• Designed a coastal-monitoring measurement system for a research association
• Developed a project website
• *Tools: CAD, Python, project management*

Numerical analysis projects
PDEs, ODEs and interpolation
• Implemented finite-difference schemes for the transport equation; analysed stability and accuracy
• Compared numerical methods: ODE solvers, Newton’s algorithm, quadrature, Runge’s phenomenon
• *Tools: Python, MATLAB/Octave*

Ocean physics projects
Waves and tides
• Studied wave reflection on a sloping plane; determined the Iribarren number experimentally
• Analysed tidal signals (HTM-NET network) using harmonic decomposition and FFT
• *Tools: Python, signal processing*

Data and materials projects
Databases, data science and materials
• Designed a relational database for the French railway network (SNCF) in SQL
• Analysed Energy Performance Certificate (DPE) data with Python
• *Tools: SQL, Python*